Name:_____ Period:_____

Entry Ticket: Minerals in Our Lives

Scientists discover that one mineral resource is becoming extremely difficult to obtain. As a result, products that depend on it may become harder to make.

1. If one mineral from the posters suddenly ran out, which one would change your daily life the most? Defend your choice.

2. Which mineral do you think would cause the biggest problem for society if we could no longer obtain it?
   a. Mineral: _____

   b. Use evidence from the posters to explain why losing this mineral would have a greater impact than losing another mineral.

3. Minerals are found in our phones, homes, and even our bodies. Why might a shortage of minerals affect people who never use a smartphone?

Name:_____ Period:_____

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Critical Minerals
IN OUR PHONES

Smartphones rely on many critical minerals. These elements give phones their strength, store and transfer electricity, and make advanced features possible.



Tantalum (Ta)
Used in small, reliable capacitors.

Camera
Rare earth elements (lens and autofocus)
Tantalum (stability)



Indium (In)
Used in touch screens for its transparency and conductivity.

Display
Indium (touch screen)
Rare earth elements (color)



Neodymium (Nd)
A rare earth element used in camera lenses and tiny magnets for autofocus.



Rare earth elements (REEs)
Such as neodymium (Nd) provide vivid colors in the display.



Circuit board
Copper (wiring)
Gold (corrosion resistance)
Silver (high conductivity)
Tantalum (capacitors)

Battery
Lithium (energy storage)
Cobalt (stability)
Nickel (energy density)
Manganese (structure)



Copper (Cu)
Conducts electricity throughout the phone.



Lithium (Li)
Allows the battery to store a large amount of energy in a small space.



Gold (Au)
Resists corrosion and ensures reliable connections.



Silver (Ag)
Offers excellent electrical conductivity in small components.

Frame and casing
Aluminum (lightweight)
Titanium (strength)



Aluminum (Al)
Makes phones lightweight and durable.



Titanium (Ti)
Provides high strength with low weight.

Where Do These Minerals Come From?

Critical minerals are mined in many parts of the world. Production is often concentrated in a few countries, which can create supply challenges.





Cobalt (Co)
Helps the battery last longer and remain stable.



Nickel (Ni)
Increases energy density, giving more power in a lighter battery.



Manganese (Mn)
Improves battery structure and safety.

Critical Minerals
IN OUR BODIES

Our bodies need minerals to grow, stay healthy and function properly. These minerals help build strong bones and teeth, carry oxygen, support muscles and nerves, and keep body systems working.



Calcium (Ca)
Builds and maintains bones and teeth. Helps with muscle function and nerve signaling.



Iron (Fe)
A key part of hemoglobin, which carries oxygen in your blood.



Sodium (Na)
Helps regulate fluid balance, nerve signaling and muscle contraction.



Chloride (Cl)
Works with sodium to maintain fluid balance and helps produce stomach acid for digestion.



Potassium (K)
Helps regulate fluid balance, nerve signals and muscle contraction, including a normal heartbeat.



Phosphorus (P)
Works with calcium to build bones and teeth. Also helps your body use energy from food and is part of DNA.



Magnesium (Mg)
Helps with muscle and nerve function, keeps a normal heartbeat and helps your body use energy from food.



Zinc (Zn)
Supports the immune system, helps wounds heal and is important for growth and cell division.



Copper (Cu)
Helps your body use iron, forms red blood cells and supports the nervous and immune systems.



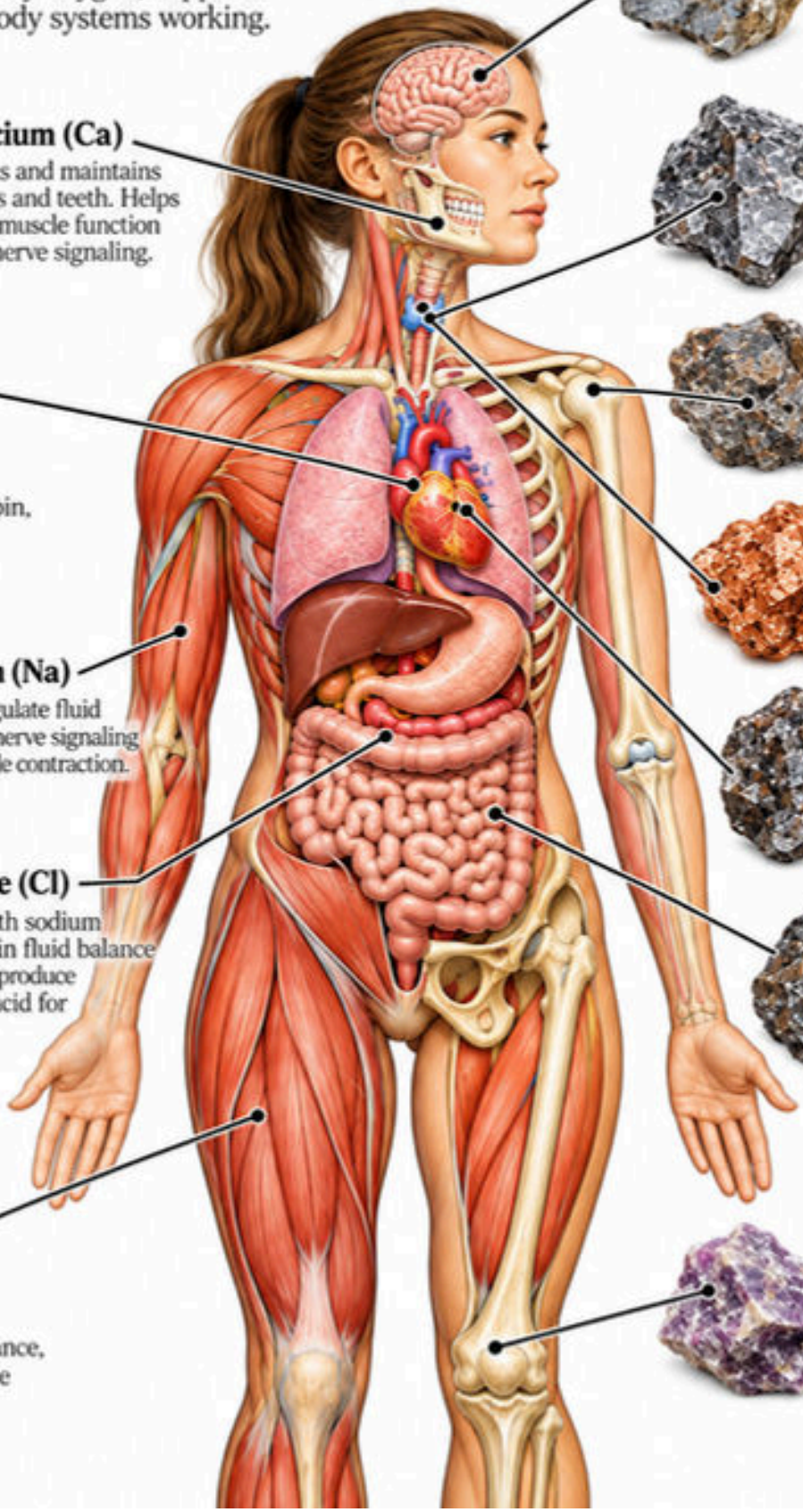
Iodine (I)
Needed to make thyroid hormones, which help control growth, metabolism and temperature.



Selenium (Se)
Helps protect cells from damage and supports a healthy immune system.



Fluoride (F)
Strengthens tooth enamel and helps prevent cavities.



Critical Minerals in Your House

Many of the things we use at home are made from or contain critical minerals.

Silicon (Si)



Used in electronics, solar panels and glass.



TV, computer, phone, windows

Aluminum (Al)



Used because it is lightweight and conducts electricity.



Refrigerator, foil, window frames, cans

Copper (Cu)



Used for electrical wiring because it conducts electricity well.



Electrical wiring, light fixtures, outlets

Iron (Fe)



Used for strong steel in buildings, appliances and tools.



Nails, screws, steel beams, tools

Calcium (Ca)



Used in cement, concrete and drywall.



Walls, floors, cement, drywall



Potassium (K)



Used in glass, ceramics and fertilizers.



Glassware, dishes, ceramic tiles

Sodium (Na)



Used in glass and cleaning products.



Glass, household cleaners, soaps

Lithium (Li)



Used in batteries for electronics and electric vehicles.



Phone, laptop, power tools, electric car battery

Nickel (Ni)



Used in stainless steel and battteries.

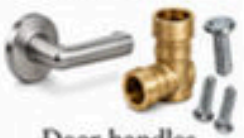


Stainless steel appliances, cookware

Zinc (Zn)



Used to prevent rust (galvanizing) and in brass.



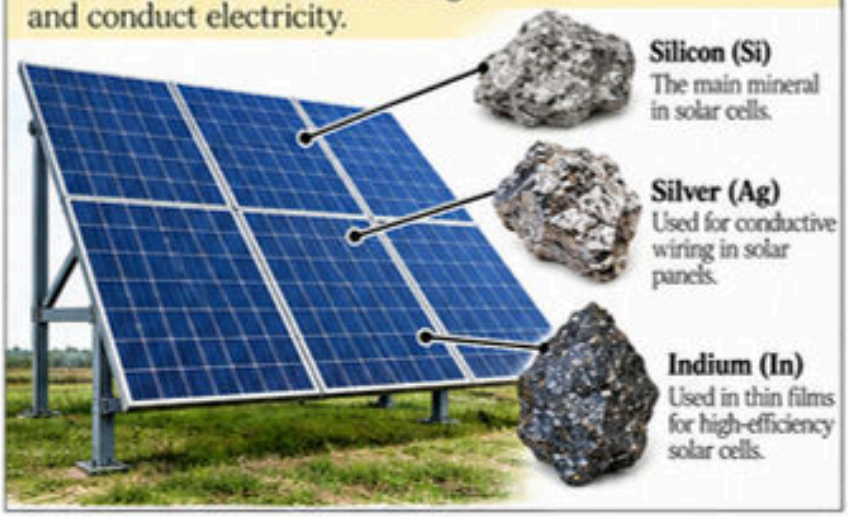
Door handles, plumbing, screws

Critical Minerals for Energy

Critical minerals are essential for producing, storing and transmitting energy. *They are used* in renewable energy technologies, power plants, batteries and the electrical grid.

Solar Panels

Use minerals that absorb sunlight and conduct electricity.



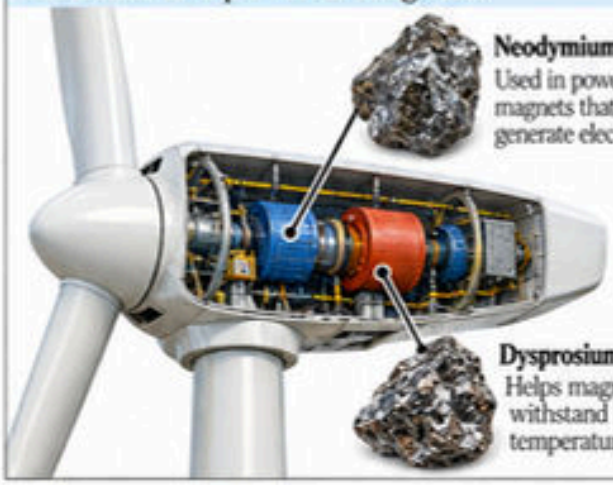
Silicon (Si)
The main mineral in solar cells.

Silver (Ag)
Used for conductive wiring in solar panels.

Indium (In)
Used in thin films for high-efficiency solar cells.

Wind Turbines

Require strong, lightweight materials and powerful magnets.



Neodymium (Nd)
Used in powerful magnets that help generate electricity.

Dysprosium (Dy)
Helps magnets withstand high temperatures.

Electric Vehicles

Rely on minerals for the battery, electric motor and other components.



Lithium (Li)
Stores a large amount of energy in the battery.

Cobalt (Co)
Improves battery stability and lifetime.

Nickel (Ni)
Increases energy density, giving more power in a lighter battery.

Manganese (Mn)
Improves battery structure and safety.

Copper (Cu)
Conducts electricity throughout the vehicle.

Power Plants

Fossil fuels, nuclear and other power plants use minerals for high-temperature materials, equipment and pollution control.



Uranium (U)
Used as fuel in nuclear power plants.

Chromium (Cr)
Used in stainless steel for high-temperature parts.

Molybdenum (Mo)
Increases strength and heat resistance in power plant components.

Transmission and Distribution

Use minerals for wires, transformers and other equipment that deliver electricity.



Copper (Cu)
Widely used in power lines because it conducts electricity well.



Aluminum (Al)
Also used for power lines because it is lightweight and conducts electricity well.